

ASSIGNMENT 9

1. What is Behaviour-Driven Development (BDD)? Explain the role of Cucumber in BDD. Write a simple example of a Gherkin feature file with Given-When-Then.

Behavior-Driven Development (BDD) is a software development approach where developers, testers, and business teams work together using simple, understandable language to describe the application's behaviour.

Role of Cucumber in BDD:

- * Cucumber is a BDD automation tool.
- * It uses the Gherkin language to write test scenarios.
- * It converts feature files into executable automated tests.
- * It improves communication between technical and non-technical team members.

2. You are testing a user login functionality with Cucumber.

Feature file

Feature: Login

Scenario: Login with valid credentials

Given User is on the Login page

When User enters username "admin"

And User enters password "admin123"

And User clicks Login button

Then User should see the Home page

Step Definition (Java)

```
import io.cucumber.java.en.Given;
```

```
public class LoginSteps {
```

```
    @Given("User is on the Login page")
```

```
    public void user_is_on_login_page() {
```

```
        System.out.println("Login page opened");
```

```
    }
```

```
}
```

3. Explain the advantages of using Cucumber with Selenium. How do Data Tables, Background, and Hooks improve test reusability? Provide examples.

Advantages of Cucumber with Selenium

- * Easy to understand using Gherkin language.
- * Improves collaboration between developers, testers, and clients.
- * Supports automated browser testing using Selenium.
- * Reusable test steps.
- * Easy maintenance of test cases.

Data Tables

Used to pass multiple sets of test data.

Background:

Given User opens the application

Example

@Before

```
public void setup() {  
    System.out.println("Browser Started");  
}
```

@After

```
public void tearDown() {  
    System.out.println("Browser Closed");  
}
```

4. What is GitHub? Explain how version control helps a testing/automation project. Write the Git commands.

GitHub is a cloud-based platform used to store, manage, and share code using Git version control.

Advantages of Version Control

- * Stores code safely.
- * Tracks all code changes.

- * Supports team collaboration.
- * Allows rollback to previous versions.
- * Maintains project history.

Git Commands

Clone a repository

git clone <https://github.com/username/repository.git>

Create a new branch

git checkout -b feature-branch

Commit changes

git add .

git commit -m "Added login automation"

Push changes to remote

Git push origin feature-branch

5. Explain the steps to configure Jenkins with Maven for running Selenium scripts.

Step 1: Install Jenkins

- * Download and install Jenkins.
- * Open Jenkins in the browser.

Step 2: Install Maven

- * Install Maven.
- * Configure Maven in Manage Jenkins → Global Tool Configuration.

Step 3: Create a Maven Job

- * Click New Item.
- * Enter the project name.
- * Select Maven Project.
- * Click OK.

Step 4: Add Dependencies in pom.xml

```
<dependencies>

    <dependency>
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-java</artifactId>
        <version>4.33.0</version>
    </dependency>

    <dependency>
        <groupId>io.cucumber</groupId>
        <artifactId>cucumber-java</artifactId>
        <version>7.20.1</version>
    </dependency>

</dependencies>
```

Step 5: Configure Build

- * Select the Git repository.
- * Specify the pom.xml file.
- * Set the Maven goal:

Step 6: Schedule the Job

- * Open Build Triggers.
- * Select Build periodically.
- * Set a schedule (for example):

Step 7: Save and Build

- * Click Save.
- * Click Build Now.
- * View the test execution results in Console Output.